

# SAFETY DATA SHEET

Creation Date 21-September-2009

Revision Date 26-March-2024

Revision Number 7

## 1. Identification

**Product Name** Trifluoroacetic acid

**Cat No. :** A14365

**CAS-No** 76-05-1  
**Synonyms** TFA; Trifluoroethanoic acid; Perfluoroacetic acid

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

### Details of the supplier of the safety data sheet

**Company**  
**Importer/Distributor**  
Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

**Emergency Telephone Number**  
For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11  
Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99  
**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

## 2. Hazard(s) identification

### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |              |
|---------------------------------------------------------|--------------|
| <b>Acute oral toxicity</b>                              | Category 4   |
| <b>Acute Inhalation Toxicity</b>                        | Category 4   |
| <b>Skin Corrosion/Irritation</b>                        | Category 1 A |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 1   |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3   |
| Target Organs - Respiratory system.                     |              |

### Label Elements

**Signal Word**  
Danger

**Hazard Statements**  
Harmful if swallowed or if inhaled

Causes severe skin burns and eye damage  
May cause respiratory irritation

**Precautionary Statements****Prevention**

Do not breathe dust/fumes/gas/mist/vapours/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Do not eat, drink or smoke when using this product  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection

**Response**

Rinse mouth  
Do NOT induce vomiting  
Wash contaminated clothing before reuse  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Immediately call a POISON CENTER/doctor

**Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Harmful to aquatic life with long lasting effects

### 3. Composition/Information on Ingredients

| Component            | CAS-No  | Weight % |
|----------------------|---------|----------|
| Trifluoroacetic acid | 76-05-1 | <=100    |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Immediate medical attention is required.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Call a physician immediately.

**Inhalation**

If not breathing, give artificial respiration. Remove from exposure, lie down. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Call a physician immediately.

**Ingestion**

Do NOT induce vomiting. Clean mouth with water. Never give anything by mouth to an

unconscious person. Call a physician immediately.

|                                        |                                                                                                                                                                                                                                                                                                 |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                           |

## 5. Fire-fighting measures

**Suitable Extinguishing Media** CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

### Specific Hazards Arising from the Chemical

Thermal decomposition can lead to release of irritating gases and vapors. The product causes burns of eyes, skin and mucous membranes.

### Hazardous Combustion Products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Hydrogen fluoride.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

### NFPA

**Health**  
3

**Flammability**  
0

**Instability**  
1

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Use personal protective equipment as required. Ensure adequate ventilation. Evacuate personnel to safe areas. Keep people away from and upwind of spill/leak.

**Environmental Precautions** Should not be released into the environment. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Do not store in metal containers. Store under an inert atmosphere. Air sensitive. Incompatible Materials. Strong oxidizing agents. Reducing Agent. Strong bases. Metals. Amines.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies.

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.  
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment****Eye Protection**

Goggles

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Butyl rubber   | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143 Acid gases filter Type E Yellow conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

Prevent product from entering drains.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

**Physical State**

Liquid

**Appearance**

Clear, Colorless

**Odor**

pungent

**Odor Threshold**

No information available

**pH**

2

100 g/L aq.sol

**Melting Point/Range**

-15 °C / 5 °F

**Softening Point**

No data available

**Boiling Point/Range**

72 °C / 161.6 °F

@ 760 mmHg

**Flash Point**

No information available

**Method -** No information available

**Flammability (liquid)**

No data available

**Flammability (solid,gas)**

Not applicable

Liquid

**Explosion Limits**

No data available

|                                                |                                                |
|------------------------------------------------|------------------------------------------------|
| <b>Autoignition Temperature</b>                | No data available                              |
| <b>Decomposition Temperature</b>               | No data available                              |
| <b>Water Solubility</b>                        | Soluble                                        |
| <b>Solubility in other solvents</b>            | No information available                       |
| <b>Partition Coefficient (n-octanol/water)</b> |                                                |
| <b>Component</b>                               | <b>log Pow</b>                                 |
| Trifluoroacetic acid                           | -2.1                                           |
| <b>Vapor Pressure</b>                          | 107 mbar @ 25 °C                               |
| <b>Density / Specific Gravity</b>              | 1.490                                          |
| <b>Bulk Density</b>                            | Not applicable                                 |
| <b>Vapor Density</b>                           | 3.9                                            |
| <b>Viscosity</b>                               | 0.813 cP at 25 °C                              |
| <b>Particle characteristics</b>                | Not applicable (liquid)                        |
| <b>Molecular Formula</b>                       | C <sub>2</sub> H F <sub>3</sub> O <sub>2</sub> |
| <b>Molecular Weight</b>                        | 114.02                                         |
| <b>Evaporation Rate</b>                        | No information available                       |

## 10. Stability and reactivity

|                                         |                                                                                        |
|-----------------------------------------|----------------------------------------------------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available                                             |
| <b>Stability</b>                        | Hygroscopic.                                                                           |
| <b>Conditions to Avoid</b>              | Incompatible products. Exposure to moist air or water. Exposure to light. Excess heat. |
| <b>Incompatible Materials</b>           | Strong oxidizing agents, Reducing Agent, Strong bases, Metals, Amines                  |
| <b>Hazardous Decomposition Products</b> | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Hydrogen fluoride             |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.                                               |
| <b>Hazardous Reactions</b>              | None under normal processing.                                                          |

## 11. Toxicological information

### Information on expected route of exposure

|                   |                                                                                                 |
|-------------------|-------------------------------------------------------------------------------------------------|
| <b>Inhalation</b> | Not an expected route of exposure.                                                              |
| <b>Ingestion</b>  | May be harmful if swallowed.                                                                    |
| <b>Eyes</b>       | Avoid contact with eyes. Corrosive to the eyes and may cause severe damage including blindness. |
| <b>Skin</b>       | Avoid contact with skin. Causes burns. Skin Corrosion/Irritation.                               |

### Acute Toxicity

#### **Product Information** **Component Information**

| Component            | LD50 Oral        | LD50 Dermal | LC50 Inhalation  |
|----------------------|------------------|-------------|------------------|
| Trifluoroacetic acid | 1200 mg/kg (Rat) | Not listed  | 10 mg/L/2h (rat) |

**Toxicologically Synergistic** No information available

#### **Products**

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | Causes severe burns by all exposure routes                                               |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component            | CAS-No  | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|----------------------|---------|------------|------------|------------|------------|------------|
| Trifluoroacetic acid | 76-05-1 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** Not mutagenic in AMES Test

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Respiratory system

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Harmful to aquatic organisms, may cause long-term adverse effects in the aquatic environment. Do not empty into drains. The product contains following substances which are hazardous for the environment.

| Component            | Freshwater Algae | Freshwater Fish                 | Microtox   | Water Flea                 |
|----------------------|------------------|---------------------------------|------------|----------------------------|
| Trifluoroacetic acid | Not listed       | Zebrafish: LC50: >1000 mg/L/96h | Not listed | daphnia: EC50: 55 mg/L/24h |

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its volatility.

| Component            | log Pow |
|----------------------|---------|
| Trifluoroacetic acid | -2.1    |

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN2699  
 Proper Shipping Name TRIFLUOROACETIC ACID  
 Hazard Class 8  
 Packing Group I

### TDG

UN-No UN2699  
 Proper Shipping Name TRIFLUOROACETIC ACID  
 Hazard Class 8  
 Packing Group I

### IATA

UN-No UN2699

|                      |                      |
|----------------------|----------------------|
| Proper Shipping Name | TRIFLUOROACETIC ACID |
| Hazard Class         | 8                    |
| Packing Group        | I                    |
| IMDG/IMO             |                      |
| UN-No                | UN2699               |
| Proper Shipping Name | TRIFLUOROACETIC ACID |
| Hazard Class         | 8                    |
| Packing Group        | I                    |

## 15. Regulatory information

All of the components in the product are on the following Inventory lists: X = listed

### International Inventories

| Component            | CAS-No  | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|----------------------|---------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Trifluoroacetic acid | 76-05-1 | X   | -    | X    | ACTIVE                                        | 200-929-3 | -      | -   |

| Component            | CAS-No  | IECSC | KECL          | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|----------------------|---------|-------|---------------|------|------|------|------|-------|-------|
| Trifluoroacetic acid | 76-05-1 | X     | KE-34233<br>X | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

| Component            | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|----------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Trifluoroacetic acid | -                                                                   | Use restricted. See entry 75. (see link for restriction details)              | -                                                                                                     |

#### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component            | CAS-No  | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|----------------------|---------|----------|------------------------------|---------------------------|--------------------------------------------|
| Trifluoroacetic acid | 76-05-1 | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component            | CAS-No  | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|----------------------|---------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Trifluoroacetic acid | 76-05-1 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

Product Safety Department  
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**Revision Summary**

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**